

Radioiodine treatment of multinodular non-toxic goitre.

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OBJECTIVE To investigate the long term effect of radioactive iodine on thyroid function and size in patients with non-toxic multinodular goitre. **DESIGN** Consecutive patients with multinodular non-toxic goitre selected for radioactive iodine treatment and followed for a minimum of 12 months (median 48 months) after an intended dose of 3.7 MBq/g thyroid tissue corrected to a 100% uptake of iodine-131 in 24 hours. **PATIENTS** 69 patients with a growing multinodular non-toxic goitre causing local compression symptoms or cosmetic inconveniences. The treatment was chosen because of a high operative risk, previous thyroidectomy, or refusal to be operated on. **MAIN OUTCOME MEASUREMENTS** Standard thyroid function variables and ultrasonically determined thyroid volume before treatment as well as 1, 2, 3, 6, and 12 months after treatment and then once a year. **RESULTS** 56 patients were treated with a single dose of ^{131}I , 12 with two doses, and one with four doses. In 45 patients treated with one dose and remaining euthyroid the median thyroid volume was reduced from 73 (interquartile range 50-106) ml to 29 (23-48) ml at 24 months in the 39 patients in whom this was measured during follow up. The median reduction was 40 (22-48) ml (60% reduction, $p < 0.0001$), half of which occurred within three months. Patients treated with two doses as well as those developing hypothyroidism and hyperthyroidism had a significant reduction in thyroid volume. Eleven patients developed hypothyroidism (cumulative five year risk 22%, 95% confidence interval 4.8% to 38.4%). Side effects were few: three cases of hyperthyroidism and two cases of radiation thyroiditis. Only one patient was dissatisfied with the result; she was referred for operation six months after treatment. **CONCLUSIONS** A substantial reduction in thyroid volume accompanied by a low incidence of hypothyroidism and few side effects makes the use of radioactive iodine an attractive alternative to surgery in selected cases of non-toxic multinodular goitre.